Multilingual Emotion Detection — Model and Training Methods

Project 3, Mining Media Data II

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1 Task and Data

We address multi-label emotion detection on the BRIGHTER corpus (SemEval-2025 Task 11) [1]: given a short text, predict any subset of six emotions (*anger, disgust, fear, joy, sadness, surprise*). We train on three languages—English (`eng`), German (`deu`) and Algerian Arabic (`arq`). English is not annotated for *disgust*, so it is scored on five labels.

Splits. Following the task protocol, the task development set is held out and evaluated exactly once. The internal split is carved from the task training set: ∼85% for training and a 15% holdout further divided into a model-selection set (early stopping) and a calibration set (threshold tuning). This keeps the development set unseen during training, selection and calibration.

2 Model

We fine-tune **XLM-RoBERTa-large** [2] as a shared multilingual encoder and attach **per-language classification heads**: each language has its own linear layer over the pooled `[CLS]` representation, with a fallback head for unseen languages. This lets the heads specialise per language while sharing cross-lingual representations in the encoder. Outputs are six independent sigmoids (multi-label).

3 Training Techniques

- **Loss:** focal loss [4] (γ=2) on top of binary cross-entropy, with per-label positive weighting (neg/pos, capped), to counter severe class imbalance (e.g. rare German *fear*/*surprise*).
- **Optimisation:** AdamW (lr 1×10^{-5}, weight decay 0.01, no decay on bias/LayerNorm), linear warmup (10%) and decay, gradient clipping (1.0), effective batch size 64 via gradient accumulation, BF16 mixed precision.
- **Sampling:** language-balanced sampling so high-resource languages do not dominate updates.
- **Model selection:** early stopping on the average per-language macro-F1 (per-language patience).
- **Data augmentation:** silver training data generated by machine translation (NLLB-200) [3] of English examples into German and Arabic, injecting minority-class signal; added to training only.
- **Threshold calibration:** per-label (and per-language) decision thresholds tuned on the held-out calibration split, with a fallback to global thresholds for labels with few positives.
- **Ensembling:** probabilities averaged across independently seeded runs to reduce variance.

4 Evaluation

The primary metric is macro-F1. Each language is scored only over the labels it annotates (English excludes *disgust*), and the overall score is the mean of the per-language macro-F1 values, reflecting cross-lingual transfer rather than letting the largest language dominate.

Configuration	Overall	`eng`	`deu`	`arq`
XLM-R large (baseline)	0.617	0.756	0.539	0.556
+ focal loss + augmentation	**0.647**	0.758	**0.625**	0.557

Table 1: Macro-F1 on the held-out development set (average of per-language macro-F1). Gains come almost entirely from German, the targeted weak language.

5 Results

Cross-lingual transfer. Performance follows resource availability. English reaches macro-F1 0.76, while German (0.62) and Algerian Arabic (0.56) trail it, illustrating the difficulty of transferring emotion detection to lower-resource languages.

Per-label F1. Averaged over languages (improved model), the easiest emotions are *anger* (0.73) and *fear* (0.66); the hardest are *joy* (0.58) and *surprise* (0.60). The single worst case is German, where *fear* and *surprise* are very rare in training and collapse to near zero under the baseline; focal loss, translation augmentation and per-language thresholds are what recover them.

	anger	disgust	fear	joy	sadness	surprise
Per-label F1	0.73	0.61	0.66	0.58	0.65	0.60

Table 2: Per-label macro-F1 (improved model, averaged over languages where the label is annotated; English excludes *disgust*).

6 Additional Local Experiments

We also ran two local model experiments on the same three languages (`eng`, `deu`, `arq`) to check whether another multilingual encoder could be competitive. These experiments are separate from the XLM-RoBERTa-large experiment above. They do **not** use per-language classification heads, focal loss, translation augmentation, language-balanced sampling, ensembling, or the task evaluation protocol. Instead, they use a shared multi-label classification head and a local calibration/evaluation split.

mDeBERTa-v3-base + LoRA. We tested `microsoft/mdeberta-v3-base`. Full fine-tuning with AdamW was too memory-heavy on the local MPS setup, so the completed run used LoRA with rank 8. This run reached macro-F1 0.5978.

mmBERT-small + LoRA. We then tested `jhu-clsp/mmBERT-small`. Full fine-tuning also exceeded local memory, so this run used LoRA on the model's attention modules. This run reached macro-F1 0.6427.

Local model	Macro-F1	Micro-F1	`arq`	`deu`	`eng`
`microsoft/mdeberta-v3-base` + LoRA	0.5978	0.5945	0.5410	0.4148	0.5323
`jhu-clsp/mmBERT-small` + LoRA	**0.6427**	**0.6411**	**0.5721**	**0.5272**	**0.5532**

Table 3: Internal validation results for our two additional model experiments. These are not task development scores and should be compared only as local results under the same simplified setup.

Local model	anger	disgust	fear	joy	sadness	surprise
mDeBERTa-LoRA	0.6800	0.6561	0.5869	0.6541	0.5182	0.4912
mmBERT-small-LoRA	**0.6940**	**0.6614**	**0.6295**	**0.6837**	**0.5441**	**0.6436**

Table 4: Per-label F1 on the internal validation split. mmBERT-small-LoRA improves over mDeBERTa-LoRA on all six labels in this setup.

The local comparison shows that mmBERT-small-LoRA is the stronger of the two additional models we tested. However, because these runs use LoRA, one seed, a shared classification head, and a different split, they do not replace the development-set result for XLM-RoBERTa-large. Their main value is to show that mmBERT-small is a better local challenger than mDeBERTa-v3-base under the same resource constraints.

7 Supplementary Prediction Evaluation

A separate prediction/evaluation script was shared for 50 translated test sentences. It uses the same XLM-RoBERTa-large model folder `output/xlmr_large_improved_seed42`. English uses the trained English head, while Spanish, Persian, Hausa and Amharic are unseen languages and use the fallback head. Therefore, these numbers evaluate fallback behaviour for unseen languages; they are not part of the BRIGHTER development-set evaluation.

Language	English	Spanish	Persian	Hausa	Amharic
Macro-F1	0.6588	0.2985	0.3019	0.2560	0.2555

Table 5: Macro-F1 on 50 translated evaluation sentences. Non-trained languages use the fallback head.

8 Limitations

- The XLM-RoBERTa-large scores are development-set results; our mDeBERTa and mmBERT scores are internal validation results and are not directly comparable to the main evaluation table.
- Our two local experiments use a shared classification head, not per-language heads.

- The local experiments use LoRA because full fine-tuning exceeded local MPS memory limits.
- The local experiments use one seed and do not include focal loss, translation augmentation, language-balanced sampling or ensembling.
- English has no *disgust* annotation in BRIGHTER. The task scoring excludes it for English; local validation scripts may handle missing labels differently.
- The translated-sentence prediction check contains only 50 sentences per language and should not be treated as a full benchmark.

9 Conclusion

The best held-out development result remains the improved XLM-RoBERTa-large model, which reaches 0.647 macro-F1. Its main gain is on German, where focal loss, augmentation and language-specific thresholding recover rare labels.

Our additional local experiments show that `jhu-clsp/mmBERT-small` + LoRA is stronger than `microsoft/mdeberta-v3-base` + LoRA under the same local setup, improving macro-F1 from 0.5978 to 0.6427. This is useful evidence for model selection, but it remains an internal validation comparison rather than a development-set result.

References

[1] Emotion Analysis Project. *BRIGHTER Corpus / SemEval-2025 Task 11*. GitHub Repository. Available at: `https://github.com/emotion-analysis-project/semeval2025-task11`

[2] Facebook AI. *XLM-RoBERTa-large*. Hugging Face Model Card. Available at: `https://huggingface.co/FacebookAI/xlm-roberta-large`

[3] Facebook AI. *NLLB-200 Distilled 600M*. Hugging Face Model Card. Available at: `https://huggingface.co/facebook/nllb-200-distilled-600M`

[4] Lin, T.-Y., Goyal, P., Girshick, R., He, K., and Dollár, P. *Focal Loss for Dense Object Detection*. Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2017. Available at: `https://arxiv.org/abs/1708.02002`